Exercises week 6

8CM10 - Introduction to Quantum Physics and Structure of Matter

Electron Configurations & Quantum Numbers

1.

   a) List the six possible sets of quantum numbers (n, l, m_l, m_s) of a $2p$ electron.
   b) Give the electron configuration of ${}^{12}_{6}C$ (carbon).
   c) Suppose we have an atom such as carbon, which has two $2p$ electrons. Ignoring the Pauli principle, how many different possible combinations of quantum numbers of the two $2p$ electrons do exist?
   d) How many of the possible combinations of part b) are eliminated by applying the Pauli principle?
   e) Suppose carbon is in an excited state with configuration $2p^1 3p^1$. Does the Pauli principle restrict the choice of quantum numbers for the electrons? How many different sets of quantum numbers are possible for the two electrons?

2. Nitrogen ($Z = 7$) has three electrons in the $2p$ level (in addition to two electrons each in the $1s$ and $2s$ levels).
   a) Consistent with the Pauli principle, what is the maximum possible value of the total m_s of all seven electrons?
   b) List the quantum numbers of the three $2p$ electrons that result in the largest total m_s.
   c) If the electrons in the $2p$ level occupy states that maximize m_s, what would be the maximum possible value for the total m_l?
   d) What would be the maximum possible total ml if the three $2p$ electrons were in states that did not maximize m_s?

3.

   a) List all elements (with $Z < 100$) having a p^3 configuration.
   b) List all elements (with $Z < 100$) having a d^7 configuration.

4. Using the wavelengths given in Figure 8.4, compute the energy difference between the $3d$ and $4d$ states in lithium; do the same for sodium. Compare those values with the corresponding $n = 4$ to $n = 3$ energy difference in hydrogen. Why is the agreement so good, considering the different values of Z?

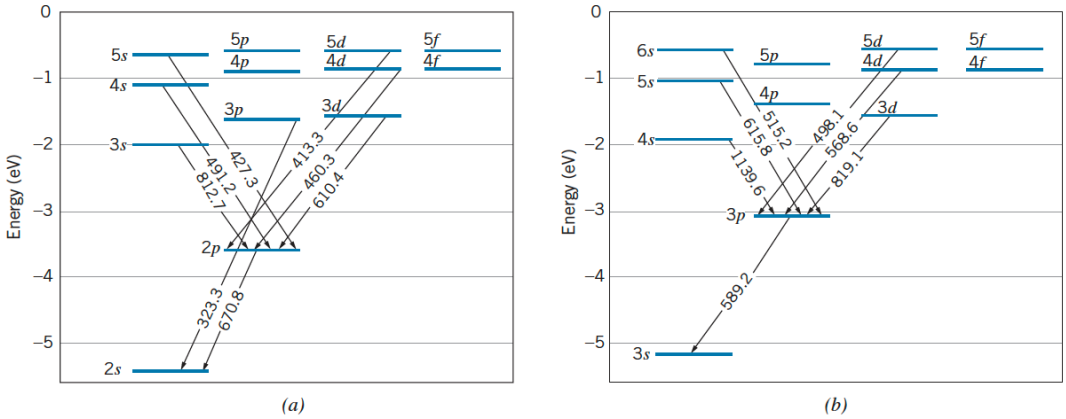


FIGURE 8.4 (a) Energy-level diagram of lithium, showing some of the transitions (labeled with wavelength in nm) in the optical region. (b) Energy-level diagram for sodium. Because of the fine-structure splitting, the $3p$ level in sodium is actually a very closely spaced pair of levels, so all transitions involving that level show two closely spaced wavelengths. The wavelength shown here is the average of the two. The fine-structure splitting is negligible for the other levels in sodium and for all of the levels in lithium.

5. The ground state of neutral beryllium ($Z = 4$) is $1s^2 2s^2$. Use the electron screening model to predict the energies of the following excited states:
   a) $1s^2 2s^1 3p^1$ (measured -2.02 eV)
   b) $1s^2 2s^1 4d^1$ (measured -0.90 eV)

X-Rays

6. The minimum wavelength of the continuous X-ray spectra from an x-ray tube is $1.24 \cdot 10^{-11}$m. What is its working potential?

7. The K-absorption edge is the frequency at which, by the photoelectric effect, a photon has enough energy to release an electron from the K-shell (i.e., n = 1) of an atom. That means that the absorption of electromagnetic radiation (X-rays) increases when the frequency exceeds the frequency of the K-absorption edge. Similarly, the L-absorption edge is the frequency at which a photon can release an electron from the L-shell (n = 2). (The two X-ray contrast media iodine and barium have ideal K-shell binding energies for absorption of X-rays, 33.2 keV and 37.4 keV, respectively, which is close to the mean energy of most diagnostic X-ray beams.)

   a) What is the relation between the frequency of the K-absorption edge and the binding energy of an electron in the K-shell of the atom?

b) The K-absorption edge of lead is at 0.14 Å. The wavelengths of the spectral lines of its K-series are 0.167 Å(K_α), 0.146 Å(K_β), and 0.142 Å(K_γ), respectively. Draw the X-ray energy level diagram for lead according to these values.
c) Calculate the minimum energy required to activate the L series and the wavelength of the L line of lead.

8. An X-ray tube with aluminum as its target can produce a continuous X-ray spectrum with a short-wave limit of 5.00 Å. Is it possible to see the aluminum characteristic K_α X-ray in its spectrum? What is the case if its short wave limit is 10 Å? The K-absorption edge of aluminum is 1.559 keV.

9. An atom in its ground level absorbs a photon with energy equal to the K-absorption edge. Does absorbing this photon ionize this atom? Explain.